(b) The marginal p.d.f. of X is

$$f_1(x) = \int_0^1 f(x,y)dy = \begin{cases} \frac{1}{3} & \text{if } 1 < x < 3, \\ \frac{1}{6} & \text{if } 6 < x < 8. \end{cases}$$

The marginal p.d.f. of Y is

$$f_2(y) = \int_1^3 \frac{1}{3}dx + \int_6^8 \frac{1}{6}dx = 1,$$

for $0 < y < 1$. The distribution of Y is the uniform distribution on the interval $[0, 1]$.

(c) The product of the two marginal p.d.f.'s is

$$f_1(x)f_2(y) = \begin{cases} \frac{1}{3} & \text{if } 1 < x < 3 \text{ and } 0 < y < 1, \\ \frac{1}{6} & \text{if } 6 < x < 8 \text{ and } 0 < y < 1, \\ 0 & \text{otherwise,} \end{cases}$$

which is the same as $f(x, y)$, hence the two random variables are independent. Although the region where $f(x, y) > 0$ is not a rectangle, it is a product set as we saw in part (a). Although it is sufficient in Theorem 3.5.6 for the region where $f(x, y) > 0$ to be a rectangle, it is necessary that the region be a product set. Technically, it is necessary that there is a version of the p.d.f. that is strictly positive on a product set. For continuous joint distributions, one can set the p.d.f. to arbitrary values on arbitrary one-dimensional curves without changing it's being a joint p.d.f.

3.6 Conditional Distributions

Commentary

When introducing conditional distributions given continuous random variables, it is important to stress that we are not conditioning on a set of 0 probability, even if the popular notation makes it appear that way. The note on page 146 can be helpful for students who understand two-variable calculus. Also, Exercise 25 in Sec. 3.11 can provide additional motivation for the idea that the conditional distribution of X given $Y = y$ is really a surrogate for the conditional distribution of X given that Y is close to y, but we don't wish to say precisely how close. Exercise 26 in Sec. 3.11 (the Borel paradox) brings home the point that conditional distributions really are not conditional on the probability 0 events such as $\{Y = y\}$.

Also, it is useful to stress that conditional distributions behave just like distributions. In particular, conditional probabilities can be calculated from conditional p.f.'s and conditional p.d.f.'s in the same way that probabilities are calculated from p.f.'s and p.d.f.'s. Also, be sure to advertise that all future concepts and theorems will have conditional versions that behave just like the marginal versions.

Solutions to Exercises

1. We begin by finding the marginal p.d.f. of Y. The set of x values for which $f(x, y) > 0$ is the interval $[-(1-y^2)^{1/2}, (1-y^2)^{1/2}]$. So, the marginal p.d.f. of Y is, for $-1 \le y \le 1$,

$$f_2(y) = \int_{-(1-y^2)^{1/2}}^{(1-y^2)^{1/2}} kx^2y^2dx = \left.\frac{ky^2}{3}x^3\right|_{x=-(1-y^2)^{1/2}}^{(1-y^2)^{1/2}} = \frac{2k}{3}y^2(1-y^2)^{3/2},$$

and 0 otherwise. The conditional p.d.f. of X given $Y = y$ is the ratio of the joint p.d.f. to the marginal p.d.f. just found.

$$g_1(x|y) = \begin{cases} \dfrac{3x^2}{2(1-y^2)^{3/2}} & \text{for } -(1-y^2)^{1/2} \le x \le (1-y^2)^{1/2}, \\ 0 & \text{otherwise.} \end{cases}$$

2. (a) We have $\Pr(\text{Junior}) = 0.04 + 0.20 + 0.09 = 0.33$. Therefore,

$$\Pr(\text{Never}|\text{Junior}) = \frac{\Pr(\text{Junior and Never})}{\Pr(\text{Junior})} = \frac{0.04}{0.33} = \frac{4}{33}.$$

(b) The only way we can use the fact that a student visited the museum three times is to classify the student as having visited more than once. We have
$\Pr(\text{More than once}) = 0.04 + 0.04 + 0.09 + 0.10 = 0.27$.
Therefore,

$$\begin{aligned}\Pr(\text{Senior}|\text{More than once}) &= \frac{\Pr(\text{Senior and More than once})}{\Pr(\text{More than once})} \\ &= \frac{0.10}{0.27} = \frac{10}{27}.\end{aligned}$$

3. The joint p.d.f. of X and Y is positive for all points inside the circle S shown in the sketch. Since the area of S is 9π and the joint p.d.f. of X and Y is constant over S, this joint p.d.f. must have the form:

$$f(x,y) = \begin{cases} \dfrac{1}{9\pi} & \text{for } (x,y) \in S, \\ 0 & \text{otherwise.} \end{cases}$$

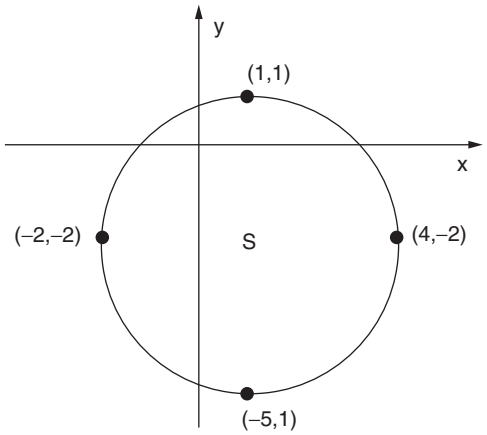


Figure S.3.31: Figure for Exercise 3 of Sec. 3.6.

It can be seen from Fig. S.3.31 that the possible values of X lie between -2 and 4. Therefore, for $-2 < x < 4$,

$$f_1(x) = \int_{-2-[9-(x-1)^2]^{1/2}}^{-2+[9-(x-1)^2]^{1/2}} \frac{1}{9\pi}\, dy = \frac{2}{9\pi}[9-(x-1)^2]^{1/2}.$$

(a) It follows that for $-2 < x < 4$ and $-2 - [9-(x-1)^2]^{1/2} < y < -2 + [9-(x-1)^2]^{1/2}$,

$$g_2(y\,|\,x) = \frac{f(x,y)}{f_1(x)} = \frac{1}{2}[9-(x-1)^2]^{-1/2}.$$

(b) When $X = 2$, it follows from part (a) that

$$g_2(y\,|\,x=2) = \begin{cases} \dfrac{1}{2\sqrt{8}} & \text{for } -2-\sqrt{8} < y < -2+\sqrt{8} \\ 0 & \text{otherwise.} \end{cases}$$

Therefore,

$$\Pr(Y > 0 \,|\, X = 2) = \int_0^{-2+\sqrt{8}} g_2(y \,|\, x = 2)\, dy = \frac{-2+\sqrt{8}}{2\sqrt{8}} = \frac{2-\sqrt{2}}{4}.$$

4. (a) For $0 \le y \le 1$, the marginal p.d.f. of y is

$$f_2(y) = \int_0^1 f(x,y)\, dx = c\left(\frac{1}{2} + y^2\right).$$

Therefore, for $0 \le x \le 1$ and $0 \le y \le 1$, the conditional p.d.f. of X given that $Y = y$ is

$$g_1(x \,|\, y) = \frac{f(x,y)}{f_2(y)} = \frac{x + y^2}{\frac{1}{2} + y^2}.$$

It should be noted that it was not necessary to evaluate the constant c in order to determine this conditional p.d.f.

(b) When $Y = 1/2$, it follows from part (a) that

$$g_1\left(x \,|\, y = \frac{1}{2}\right) = \begin{cases} \frac{4}{3}\left(x + \frac{1}{4}\right) & \text{for } 0 \le x \le 1, \\ 0 & \text{otherwise.} \end{cases}$$

Therefore,

$$\Pr\left(X < \frac{1}{2} \,|\, Y = \frac{1}{2}\right) = \int_0^{\frac{1}{2}} g_1\left(x \,|\, y = \frac{1}{2}\right) dx = \frac{1}{3}.$$

5. (a) The joint p.d.f. $f(x,y)$ is given by Eq. (3.6.15) and the marginal p.d.f. $f_2(y)$ was also given in Example 3.6.10. Hence, for $0 < y < 1$ and $0 < x < y$, we have

$$g_1(x \,|\, y) = \frac{f(x,y)}{f_2(y)} = \frac{-1}{(1-x)\ \log(1-y)}.$$

(b) When $Y = 3/4$, it follows from part (a) that

$$g_1\left(x \,|\, y = \frac{3}{4}\right) = \begin{cases} \frac{1}{(1-x)\log\ 4} & \text{for } 0 < x < \frac{3}{4}, \\ 0 & \text{otherwise.} \end{cases}$$

Therefore,

$$\Pr\left(X > \frac{1}{2} \,|\, Y = \frac{3}{4}\right) = \int_{1/2}^{3/4} g_1\left(x \,|\, y = \frac{3}{4}\right) dx = \frac{\log\ 4 - \log\ 2}{\log\ 4} = \frac{1}{2}.$$

6. Since $f(x,y) = 0$ outside a rectangle with sides parallel to the x and y axes and since $f(x,y)$ can be factored as in Eq. (3.5.7), with $g_1(x) = c\sin(x)$ and $g_2(y) = 1$, it follows that X and Y are independent random variables. Furthermore, for $0 \le y \le 3$, the marginal p.d.f. $f_2(y)$ must be proportional to $g_2(y)$. In other words, $f_2(y)$ must be constant for $0 \le y \le 3$. Hence, Y has the uniform distribution on the interval $[0, 3]$ and

$$f_2(y) = \begin{cases} \frac{1}{3} & \text{for } 0 \le y \le 3, \\ 0 & \text{otherwise.} \end{cases}$$

(a) Since X and Y are independent, the conditional p.d.f. of Y for any given value of X is the same as the marginal p.d.f. $f_2(y)$.

(b) Since X and Y are independent,

$$\Pr(1 < Y < 2 \,|\, X = 0.73) = \Pr(1 < Y < 2) = \int_1^2 f_2(y)\, dy = \frac{1}{3}.$$

7. The joint p.d.f. of X and Y is positive inside the triangle S shown in Fig. S.3.32. It is seen from Fig. S.3.32 that the possible values of X lie between 0 and 2. Hence, for $0 < x < 2$,

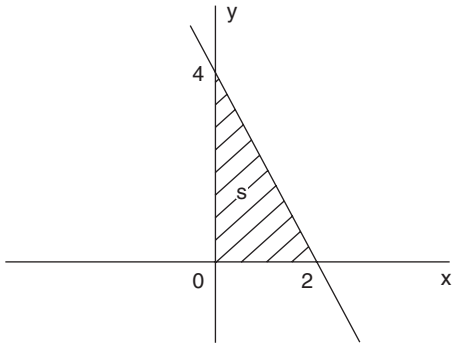


Figure S.3.32: Figure for Exercise 7 of Sec. 3.6.

$$f_1(x) = \int_0^{4-2x} f(x, y)\, dy = \frac{3}{8}(x-2)^2.$$

(a) It follows that for $0 < x < 2$ and $0 < y < 4 - 2x$,

$$g_2(y \,|\, x) = \frac{f(x, y)}{f_1(x)} = \frac{4 - 2x - y}{2(x-2)^2}.$$

(b) When $X = 1/2$, it follows from part (a) that

$$g_2\left(y \,|\, x = \frac{1}{2}\right) = \begin{cases} \frac{2}{9}(3-y) & \text{for } 0 < y < 3, \\ 0 & \text{otherwise.} \end{cases}$$

Therefore,

$$\Pr\left(Y \geq 2 \,|\, X = \frac{1}{2}\right) = \int_2^3 g_2\left(y \,|\, x = \frac{1}{2}\right) dy = \frac{1}{9}.$$

8. (a) The answer is

$$\int_0^1 \int_{0.8}^1 f(x, y)\, dx\, dy = 0.264.$$

(b) For $0 \leq y \leq 1$, the marginal p.d.f. of Y is

$$f_2(y) = \int_0^1 f(x, y)\, dx = \frac{2}{5}(1 + 3y).$$

Hence, for $0 \leq x \leq 1$ and $0 \leq y \leq 1$,

$$g_1(x \,|\, y) = \frac{2x + 3y}{1 + 3y}.$$

When $Y = 0.3$, it follows that

$$g_1(x \,|\, y = 0.3) = \frac{2x + 0.9}{1.9} \quad \text{for } 0 \leq x \leq 1.$$

Hence,

$$\Pr(X > 0.8 \,|\, Y = 0.3) = \int_{0.8}^{1} g_1(x \,|\, y = 0.3) dx = 0.284.$$

(c) For $0 \le x \le 1$, the marginal p.d.f. of X is

$$f_1(x) = \int_0^1 f(x, y)\, dy = \frac{2}{5}\left(2x + \frac{3}{2}\right).$$

Hence, for $0 \le x \le 1$ and $0 \le y \le 1$,

$$g_2(y \,|\, x) = \frac{2x + 3y}{2x + \frac{3}{2}}.$$

When $X = 0.3$, it follows that

$$g_2(y \,|\, x = 0.3) = \frac{0.6 + 3y}{2.1} \quad \text{for } 0 \le y \le 1.$$

Hence,

$$\Pr(Y > 0.8 \,|\, X = 0.3) = \int_{0.8}^{1} g_2(y \,|\, x = 0.3)\, dy = 0.314.$$

9. Let Y denote the instrument that is chosen. Then $\Pr(Y = 1) = \Pr(Y = 2) = 1/2$. In this exercise the distribution of X is continuous and the distribution of Y is discrete. Hence, the joint distribution of X and Y is a mixed distribution, as described in Sec. 3.4. In this case, the joint p.f./p.d.f. of X and Y is as follows:

$$f(x, y) = \begin{cases} \frac{1}{2} \cdot 2x = x & \text{for } \ y = 1 \text{ and } \ 0 < x < 1, \\ \frac{1}{2} \cdot 3x^2 = \frac{3}{2}x^2 & \text{for } \ y = 2 \text{ and } \ 0 < x < 1, \\ 0 & \text{otherwise.} \end{cases}$$

(a) It follows that for $0 < x < 1$,

$$f_1(x) = \sum_{y=1}^{2} f(x, y) = x + \frac{3}{2}x^2,$$

and $f_1(x) = 0$ otherwise.

(b) For $y = 1, 2$ and $0 < x < 1$, we have

$$\Pr(Y = y \,|\, X = x) = g_2(y \,|\, x) = \frac{f(x, y)}{f_1(x)}.$$

Hence,

$$\Pr\left(Y = 1 \,|\, X = \frac{1}{4}\right) = \frac{f\left(\frac{1}{4}, 1\right)}{f_1\left(\frac{1}{4}\right)} = \frac{\frac{1}{4}}{\frac{1}{4} + \frac{3}{2} \cdot \frac{1}{16}} = \frac{8}{11}.$$

10. Let $Y = 1$ if a head is obtained when the coin is tossed and let $Y = 0$ if a tail is obtained. Then $\Pr(Y = 1 \,|\, X = x) = x$ and $\Pr(Y = 0 \,|\, X = x) = 1 - x$. In this exercise, the distribution of X is

continuous and the distribution of Y is discrete. Hence, the joint distribution of X and Y is a mixed distribution as described in Sec. 3.4. The conditional p.f. of Y given that $X = x$ is

$$g_2(y \,|\, x) = \begin{cases} x & \text{for } y = 1, \\ 1 - x & \text{for } y = 0, \\ 0 & \text{otherwise.} \end{cases}$$

The marginal p.d.f. $f_1(x)$ of X is given in the exercise, and the joint p.f./p.d.f. of X and Y is $f(x, y) = f_1(x)g_2(y \,|\, x)$. Thus, we have

$$f(x, y) = \begin{cases} 6x^2(1 - x) & \text{for } 0 < x < 1 \text{ and } y = 1, \\ 6x(1 - x)^2 & \text{for } 0 < x < 1 \text{ and } y = 0, \\ 0 & \text{otherwise.} \end{cases}$$

Furthermore, for $y = 0, 1$,

$$\Pr(Y = y) = f_2(y) = \int_0^1 f(x, y)\, dx.$$

Hence,

$$\Pr(Y = 1) = \int_0^1 6x^2(1 - x)\, dx = \int_0^1 (6x^2 - 6x^3)\, dx = \frac{1}{2}.$$

(This result could also have been derived by noting that the p.d.f. $f_1(x)$ is symmetric about the point $x = 1/2$.)

It now follows that the conditional p.d.f. of X given that $Y = 1$ is, for $0 < x < 1$,

$$g_1(x \,|\, y = 1) = \frac{f(x, 1)}{\Pr(Y = 1)} = \frac{6x^2(1 - x)}{1/2} = 12x^2(1 - x).$$

11. Let F_2 be the c.d.f. of Y. Since f_2 is continuous at both y_0 and y_1, we can write, for $i = 0, 1$,

$$\Pr(Y \in A_i) = F_2(y_i + \epsilon) - F_2(y_i - \epsilon) = 2\epsilon f_2(y_i'),$$

where y_i' is within ϵ of y_i. This last equation follows from the mean value theorem of calculus. So

$$\frac{\Pr(Y \in A_0)}{\Pr(Y \in A_1)} = \frac{f_2(y_0')}{f_2(y_1')}. \tag{S.3.1}$$

Since f_2 is continuous, $\lim_{\epsilon \to 0} f_2(y_i') = f_2(y_i)$, and the limit of (S.3.1) is $0/f_2(y_1) = 0$.

12. (a) The joint p.f./p.d.f. of X and Y is the product $f_2(y)g_1(x|y)$.

$$f(x, y) = \begin{cases} (2y)^x \exp(-3y)/x! & \text{if } y > 0 \text{ and } x = 0, 1, \ldots, \\ 0 & \text{otherwise.} \end{cases}$$

The marginal p.f. of X is obtained by integrating over y.

$$f_1(x) = \int_0^\infty \frac{(2y)^x}{x!} \exp(-3y) dy = \frac{1}{3}\left(\frac{2}{3}\right)^x,$$

for $x = 0, 1, \ldots$.

(b) The conditional p.d.f. of Y given $X = 0$ is the ratio of the joint p.f./p.d.f. to $f_1(0)$.

$$g_2(y|0) = \frac{(2y)^0 \exp(-3y)/0!}{(1/3)(2/3)^0} = 3\exp(-3y),$$

for $y > 0$.

(c) The conditional p.d.f. of Y given $X = 1$ is the ratio of the joint p.f./p.d.f. to $f_1(1)$.

$$g_2(y|1) = \frac{(2y)^1 \exp(-3y)/1!}{(1/3)(2/3)^1} = 9y\exp(-3y),$$

for $y > 0$.

(d) The ratio of the two conditional p.d.f.'s is

$$\frac{g_2(y|1)}{g_2(y|0)} = \frac{9y\exp(-3y)}{3\exp(-3y)} = 3y.$$

The ratio is greater than 1 if $y > 1/3$. This corresponds to the intuition that if we observe more calls, then we should think the rate is higher.

13. There are four different treatments on which we are asked to condition. The marginal p.f. of treatment Y is given in the bottom row of Table 3.6 in the text. The conditional p.f. of response given each treatment is the ratio of the two rows above that to the bottom row:

$$g_1(x|1) = \begin{cases} \frac{0.120}{0.267} = 0.4494 & \text{if } x = 0, \\ \frac{0.147}{0.267} = 0.5506 & \text{if } x = 1. \end{cases}$$

$$g_1(x|2) = \begin{cases} \frac{0.087}{0.253} = 0.3439 & \text{if } x = 0, \\ \frac{0.166}{0.253} = 0.6561 & \text{if } x = 1. \end{cases}$$

$$g_1(x|3) = \begin{cases} \frac{0.146}{0.253} = 0.5771 & \text{if } x = 0, \\ \frac{0.107}{0.253} = 0.4229 & \text{if } x = 1. \end{cases}$$

$$g_1(x|4) = \begin{cases} \frac{0.160}{0.227} = 0.7048 & \text{if } x = 0, \\ \frac{0.067}{0.227} = 0.2952 & \text{if } x = 1. \end{cases}$$

The fourth one looks quite different from the others, especially from the second.

3.7 Multivariate Distributions

Commentary

The material around Definition 3.7.8 and Example 3.7.8 reintroduces the concept of conditionally independent random variables. This concept is important in Bayesian inference, but outside of Bayesian inference, it generally appears only in more advanced applications such as expert systems and latent variable models. If an instructor is going to forego all discussion of Bayesian inference then this material (and Exercises 13 and 14) could be skipped.

Solutions to Exercises

1. (a) We have

$$\int_0^1 \int_0^1 \int_0^1 f(x_1, d_2, x_3)\, dx_1\, dx_2\, dx_3 = 3c.$$

Since the value of this integral must be equal to 1, it follows that $c = 1/3$.